

Annuity Due

- This cash flow pattern is referred to as an **annuity due**. It differs from a standard annuity in that the first of the N payments occurs at time 0 (now) rather than at the end of the first time period.

A car loan requires 30 monthly payments of \$199.00, starting *today*. At an annual rate of 12 percent compounded monthly, how much money is being lent?

where $A = 199$, $i = 12\%/12 = 1\%$, and $N = 29$.

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$$\begin{aligned} P &= 199 + 199(P/A, 1\%, 29) \\ &= 199 + 199(25.066) \\ &= 199 + 4988.13 \\ &= 5187.13 \end{aligned}$$



Practice question

Consider three investment plans at an annual interest rate of 9.38% (Figure 3.22):

- **Investor A.** Invest \$2,000 per year for the first 10 years of your career. At the end of 10 years, make no further investments, but reinvest the amount accumulated at the end of 10 years for the next 31 years.
- **Investor B.** Do nothing for the first 10 years. Then start investing \$2,000 per year for the next 31 years.
- **Investor C.** Invest \$2,000 per year for the entire 41 years.

Note that all investments are made at the *beginning* of each year; the first deposit will be made at the beginning of age 25 ($n = 0$), and you want to calculate the balance at the age of 65 ($n = 41$).

Practice question



To help you reach a \$5,000 goal five years from now, your father offers to give you \$500 now. You plan to get a part-time job and make five additional deposits, one at the end of each year. (The first deposit is made at the end of the first year.) If all your money is deposited in a bank that pays 7% interest, how large must your annual deposit be?

Uniform Gradient Cash flow Series

- An **arithmetic gradient** series is a cash flow series that either increases or decreases by a **constant amount** each period. The amount of change is called the **gradient**.
- Assume a manufacturing engineer predicts that the cost of maintaining a robot will increase by \$5000 per year until the machine is retired. The cash flow series of maintenance costs involves a constant gradient, which is \$5000 per year.

Uniform Gradient Cash flow Series

- For example, if you purchase a used car with a 1-year warranty, you might expect to pay the gasoline and insurance costs during the first year of operation. Assume these cost \$2500; that is, \$2500 is the base amount. After the first year, you absorb the cost of repairs, which can be expected to increase each year. If you estimate that total costs will increase by \$200 each year, the amount the second year is \$2700, the third \$2900, and so on to year n , when the total cost is $2500 + (n - 1)200$. Note that the gradient (\$200) is first observed between year 1 and year 2, and the base amount (\$2500 in year 1) is not equal to the gradient.
- Define the symbols G for gradient and A for cash flow in year n as follows.

G = constant arithmetic change in cash flows from one time period to the next; G may be positive or negative.

$$CF_n = \text{base amount} + (n - 1)G$$

[2.18]